

CHILDRESS, TX, USA

WMO: 723660

Lat: 34.427N

Lon: 100.283W

Elev: 595

StdP: 94.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23007

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-6.1	-16.9	0.9	-0.9	-14.6	1.1	0.4	14.0	10.0	12.4	9.8	4.1	10	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	39.1	21.3	37.6	21.7	36.3	21.6	24.7	33.2	24.0	32.5	23.4	31.9	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	18.4	28.2	21.7	17.6	27.4	21.1	17.0	26.8	78.1	33.3	75.3	32.5	72.6	31.8	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	11.0	9.5	-12.3	41.4	2.6	2.3	-14.2	43.1	-15.7	44.4	-17.2	45.7	-19.1	47.5		
			-13.2	25.6	2.4	1.2	-14.9	26.5	-16.3	27.2	-17.6	27.8	-19.3	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	5.6	7.4	12.3	17.0	22.0	26.7	28.6	28.0	23.7	17.5	11.3	6.1
	DBStd	9.46	5.46	6.10	5.78	5.02	4.61	3.19	2.59	2.68	3.74	4.74	5.13	5.09
	HDD10.0	505	154	110	43	7	1	0	0	0	0	6	45	140
	HDD18.3	1676	393	308	200	82	18	0	0	0	6	73	216	380
	CDD10.0	3144	19	37	113	218	373	501	576	558	412	237	83	19
	CDD18.3	1272	0	1	12	42	132	251	317	299	167	46	4	0
	CDH23.3	15602	14	54	209	583	1551	3029	4146	3647	1753	516	90	11
	CDH26.7	8229	1	13	64	238	765	1653	2393	2058	838	186	19	1
Wind	WSAvg	4.7	4.6	4.9	5.3	5.9	5.4	5.3	4.4	3.9	4.0	4.4	4.5	4.4
Precipitation	PrecAvg	512	17	23	34	48	84	84	37	59	64	41	28	17
	PrecMax	870	83	61	126	186	207	210	133	184	171	108	109	94
	PrecMin	320	0	0	0	2	0	10	1	0	0	0	0	0
	PrecStd	119	20	19	32	41	55	58	33	40	52	36	28	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	28.7	31.7	35.2	39.0	41.6	40.8	40.7	37.9	33.7	29.2	25.0
		MCWB	11.8	13.1	15.3	15.3	18.3	20.5	21.7	21.3	21.3	18.4	14.8	12.2
	2%	DB	21.6	24.6	28.3	32.3	36.1	38.2	38.7	38.5	35.5	31.1	26.0	20.9
		MCWB	10.0	12.0	14.2	15.7	19.1	21.6	21.9	21.9	21.2	18.5	14.2	10.5
	5%	DB	18.0	21.1	25.6	29.5	33.6	36.1	37.4	37.0	33.6	28.7	23.0	17.6
		MCWB	8.6	10.7	13.8	15.7	19.4	21.5	22.1	22.2	21.0	17.9	13.1	9.1
	10%	DB	15.1	17.5	22.5	26.6	31.1	34.1	36.0	35.3	31.6	26.3	20.3	14.8
		MCWB	7.2	9.1	12.8	15.3	18.9	21.7	22.1	22.2	20.5	16.8	12.2	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.7	18.6	20.5	23.5	25.5	25.6	25.6	24.3	22.4	17.9	15.1
		MCDB	21.3	21.7	26.2	27.7	31.4	34.0	34.1	33.4	32.4	28.3	23.2	19.2
	2%	WB	11.1	13.3	17.2	19.3	22.5	24.4	24.6	24.5	23.2	21.0	16.2	11.9
		MCDB	19.2	21.5	23.6	25.7	30.1	32.8	33.3	33.0	30.9	27.0	21.7	18.8
	5%	WB	9.4	11.6	15.6	18.1	21.5	23.6	23.9	23.8	22.4	19.7	14.8	9.9
		MCDB	17.1	19.4	21.9	24.6	29.1	32.0	32.6	32.4	29.7	25.0	20.5	17.0
	10%	WB	7.7	9.8	13.9	16.8	20.6	22.8	23.2	23.1	21.6	18.4	13.3	8.1
		MCDB	14.4	17.2	20.9	23.4	28.2	31.2	32.1	31.7	28.6	23.6	19.3	14.2
Mean Daily Temperature Range	5% DB	MDBR	13.9	13.9	14.5	14.6	13.4	12.8	12.9	12.9	13.0	13.7	13.8	13.0
		MCDBR	19.3	19.9	18.9	18.7	17.1	15.6	14.9	15.2	15.6	16.9	18.4	17.8
		MCWBR	10.4	10.2	8.7	7.8	6.6	5.2	4.5	4.9	5.5	7.3	9.0	9.6
	5% WB	MCDBR	18.0	17.7	15.3	14.9	14.0	13.0	12.7	12.8	12.5	12.8	14.4	16.5
		MCWBR	10.2	10.2	8.1	7.7	6.4	5.3	4.7	4.8	5.1	6.7	8.1	9.6
Clear-Sky Solar Irradiance	taub	0.276	0.287	0.320	0.355	0.361	0.399	0.408	0.391	0.361	0.343	0.308	0.281	
	taud	2.558	2.535	2.450	2.368	2.424	2.349	2.378	2.377	2.440	2.464	2.499	2.547	
	Ebn at Noon	948	970	956	931	923	883	873	885	900	890	896	919	
	Edh at Noon	78	90	106	121	116	125	121	119	106	95	83	75	
All-Sky Solar Radiation	RadAvg	3.15	3.83	5.08	6.15	6.67	7.22	7.09	6.42	5.38	4.18	3.27	2.71	
	RadStd	0.29	0.46	0.31	0.45	0.65	0.53	0.35	0.34	0.32	0.52	0.37	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	N/A

Nomenclature: See separate page